



KHALISHAH NADHIRAH ABU BAKAR

Lead Data Scientist with 9+ years of experience driving end-to-end ML and analytics transformation across public and private sectors. Proven track record in building and scaling production-grade machine learning systems, leading cross-functional teams, and translating complex models into high-impact business decisions.

Specialises in fraud analytics, explainable AI, and large-scale data platforms on AWS.

📞 khalishahnadhirah@gmail.com

🌐 [linkedin.com/in/khalishah-nadhirah-abu-bakar](https://www.linkedin.com/in/khalishah-nadhirah-abu-bakar)

📍 Singaporean, open for relocation

WORK EXPERIENCE

LEAD DATA SCIENTIST | 04.2026 - Present | Singapore

SENIOR MANAGER, DATA SCIENTIST | 03.2023 - 03.2026 | Singapore

Skillsfuture Singapore (SSG)

- Data Science Lead (Fraud Analytics Models (FAMs))
 - Led a cross-functional team of 5 (Data Scientists, ML Engineer, Data Engineer) delivering a suite of enterprise fraud analytics models end-to-end
 - Transformed legacy black-box models into fully explainable models, achieving 100% transparency and improving business users' trust
 - Designed and implemented model monitoring & explainability frameworks, reducing maintenance effort by 50%
 - Migrated ML infrastructure from on-prem to AWS, reducing costs by 30% and improving scalability and deployment speed
 - Established production deployment best practices (shadow testing, canary releases, version control), reducing model refresh cycles by 50%
 - Drove alignment across data, operations, and policy teams, achieving 95% on-time delivery of critical analytics initiatives
- Data Science Lead (POCs & Strategy)
 - Led a team of 7 delivering rapid analytics and AI POCs (policy research, network analysis, risk detection, KPI tracking) within 3-month cycles
 - Translated business and policy problems into scalable data science solutions, increasing cross-division impact of analytics
 - Trained 30+ stakeholders in data literacy, improving adoption and self-service analytics capabilities
 - Mentored 5 junior data scientists, improving team output quality by 20%

DATA SCIENTIST | 11.2020 - 03.2023 | Barcelona, Spain

benshi.ai

- Developed standardised logics for metrics, KPIs, and features, enabling the processing of millions of event data points to deliver real-time insights and data-driven decision-making for stakeholders, through our own in-house AI-as-a-Service (AIAas) platform.
- Led a cross-functional team predicting customer attrition for a midwifery management application, saving 100+ accounts within six months through advanced modelling in reinforcement learning and adaptive interventions for early risk detection.
- Researched and assessed advanced survival models (Dynamic DeepHit, Deep Survival Machines, etc) to enhance the engine behind our AIAas platform.
- Built and optimised demand forecasting models for supply chain datasets, preventing stock shortages and improving inventory optimisation for stakeholders.
- Designed and executed experimentation frameworks (A/B testing, Multi-Armed Bandits), increasing daily active users by 30% through data-driven insights into user behaviour.
- Automated experimentation workflows, saving 5+ hours/week and increasing team efficiency
- Led the onboarding process for new team members, reducing onboarding time from 4 weeks to 2 weeks, ensuring faster integration into active projects.
- Collaborated with product and engineering teams to deploy real-time analytics dashboards for decision-making

DATA SCIENTIST | 07.2018 - 11.2020 | Singapore

Infocepts, Contracted for DFS Group

- Contributed to the development of an enterprise Artificial Intelligence (AI) platform for DFS, supporting organisation-wide data-driven decision-making
- Engineered feature-rich data assets for model prototyping, enabling the development of robust machine learning models.
- Built and deployed various prototype models:
 - **Propensity Models** for Brands, delivering a 40% uplift in performance compared to business-as-usual, significantly improving targeted marketing.
 - **Customer Segmentation Models** using Recency, Frequency, and Monetary (RFM) analysis, facilitating personalised marketing strategies and customer engagement.
 - **Demand Forecasting Models** using time series analysis, optimising inventory and sales forecasting, enhancing supply chain efficiency.
 - **Survival Regression** on brands, providing deeper insights into brand lifecycles and retention patterns.
- Led the development of an anomaly detection model using Autoencoders and classification techniques to identify professional shoppers, achieving a 50% improvement in customer targeting accuracy and reducing fraudulent activities.

CONTENT DEVELOPER | 08.2017 - 06.2018 | Singapore

Tueetor

- Produced 20-25 graphics, 4 blogs, and 3 corporate videos monthly for clients, as well as 2 in-house engagement videos, ensuring timely delivery of high-quality content.
- Contributed to creative ideation for social media content, improving brand visibility across multiple platforms.
- Collaborated with clients and internal teams to align content with business goals and brand identity.
- Managed end-to-end production of videos and graphics, driving engagement and ensuring seamless execution of content.

STATISTICAL RESEARCHER | 04.2016 - 07.2017 | Singapore

Muslim Healthcare Professionals Association (MHPA)

- Analysed medical datasets from a clinical program at Singapore General Hospital, examining the impact of education and medication adjustments on the risk of diabetes complications during the fasting month.
- Co-authored a research paper accepted for publication in SAGE Journals, Therapeutic Advances in Endocrinology and Metabolism.



KHALISHAH NADHIRAH ABU BAKAR

Lead Data Scientist with 9+ years of experience driving end-to-end AI and analytics transformation across public and private sectors. Proven track record in building and scaling production-grade machine learning systems, leading cross-functional teams, and translating complex models into high-impact business decisions.

Specialises in fraud analytics, explainable AI, and large-scale data platforms on AWS.

📞 khalishahnadhirah@gmail.com

🌐 linkedin.com/in/khalishah-nadhirah-abu-bakar

📍 Singaporean, open for relocation

EDUCATION

MASTER OF SCIENCE IN URBAN SCIENCE, POLICY AND PLANNING | 09.2020 - 09.2021
SINGAPORE UNIVERSITY OF TECHNOLOGY & DESIGN (SUTD)
MSc (Summa Cum Laude) in Urban Science, Policy & Planning (Data Science Specialisation)

BACHELOR OF SCIENCE, MAJOR IN STATISTICS | 08.2013 - 02.2017
NATIONAL UNIVERSITY OF SINGAPORE (NUS)
BSc (Pass with Merit), Major in Statistics with Biostatistics Specialisation

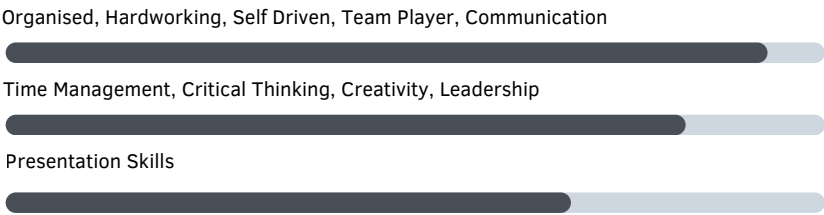
STUDENT EXCHANGE PROGRAMME | 01.2015 - 06.2015
FREIBURG UNIVERSITY (SPRACHLEHRINSTITUT)
German Language Immersion Programme

PRE-UNIVERSITY GRADUATE | 01.2011 - 12.2012
TAMPINES JUNIOR COLLEGE
Advanced Level (A-Level) Singapore-Cambridge General Certificate of Education

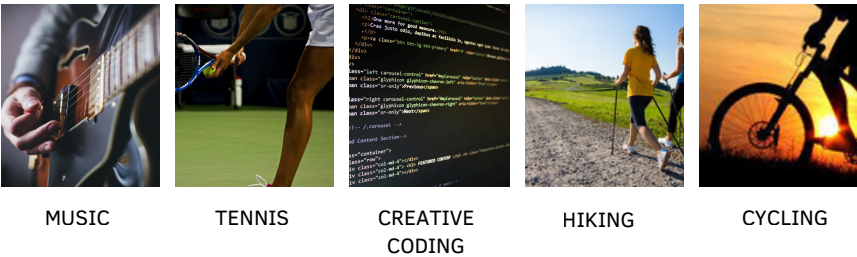
VOLUNTEER EXPERIENCE

RESEARCH & DATA LEAD | *OPENBOOKSG* | 03.2019 - PRESENT
DATA & FINANCE DIRECTOR | *CONNECTUS* | 04.2019 - 08.2020
DATA TEAM MEMBER | *YOUNG MENDAKI CLUB* | 03.2019 - 10.2019
DATA SCIENCE VOLUNTEER | *THE CODETTE PROJECT* | 08.2018 - PRESENT
WTA VOLUNTEER | *WOMEN'S TENNIS ASSOCIATION* | 10.2017 & 10.2018
DATA ANALYTICS MENTOR | *MADRASAH AL-ARABIAH AL-ISLAMIAH* | 07.2017
TUTOR | *GLOBAL PASSPORT, AIESEC HANOI* | 01.2016 - 02.2016
ASSISTANT HONORARY TREASURER | *NUS MUSLIM SOCIETY* | 12.2013 - 07.2015

COMPETENCIES



INTERESTS



SKILLS

● ● ● ● ● VERY PROFICIENT

Programming Languages
Python, R, Pyspark, SparkR, SQL

Cloud Services
AWS (Services relevant to Data Science and Machine Learning), Snowflake, Databricks

Data, ML & Analytics Tools
Alteryx, Tableau

Project Management Tools
Microsoft Office Suite, Notion, Github, Jira, Confluence

Creative & Design Tools
Adobe Photoshop

● ● ● ● ● PROFICIENT

Data, ML & Analytics Tools
Aginity Netezza Workbench, Hive, MLflow, Microstrategy, Microsoft Power BI, neo4j, Talend

Project Management Tools
Overleaf

Creative & Design Tools
Adobe Premiere Pro

● ● ● ● ● LEAST PROFICIENT

Programming Languages & Frameworks
C, Tensorflow, Octave

Data Science, ML & Analytics Tools
SAS, SPSS, H2O.ai, DataRobot

Creative & Design Tools
Adobe Illustrator

LANGUAGES

